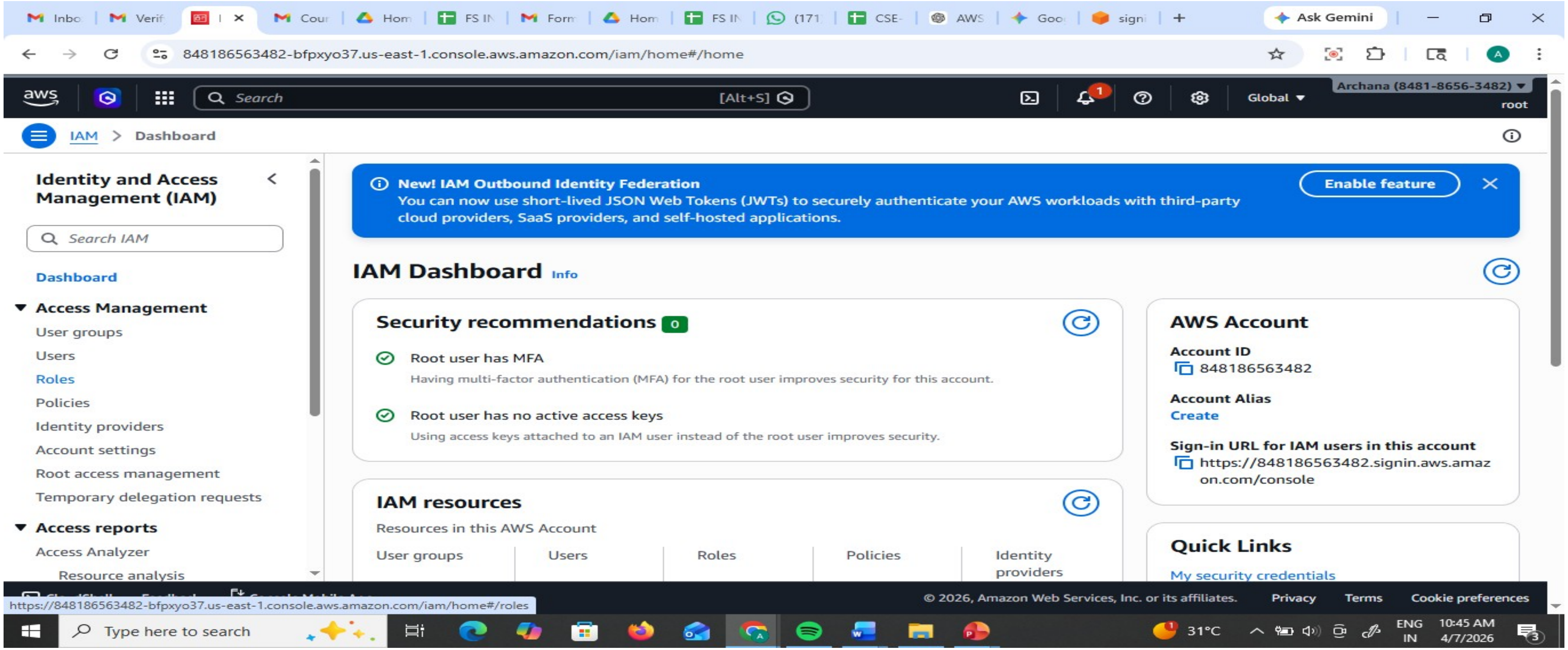
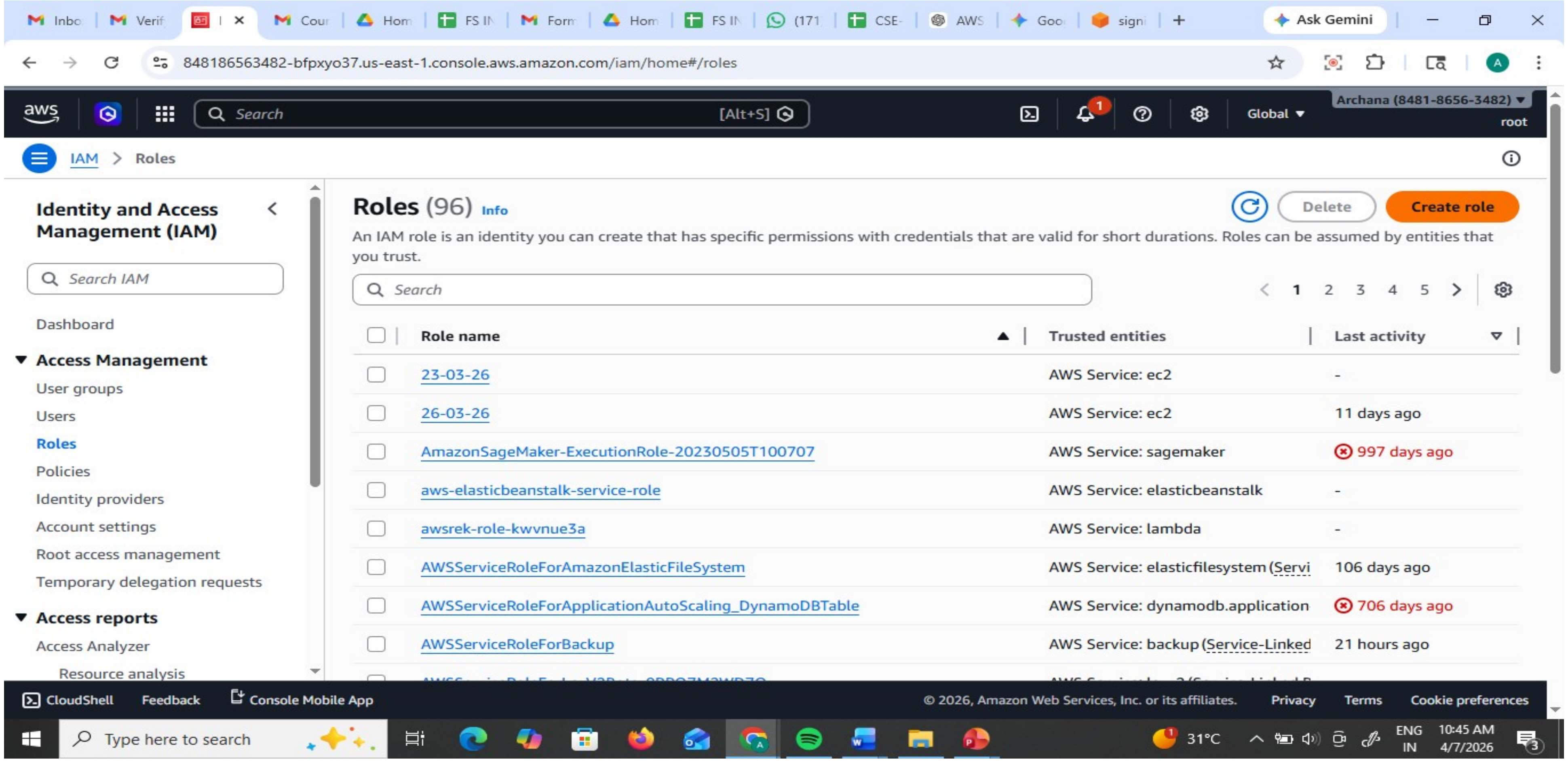
Roles

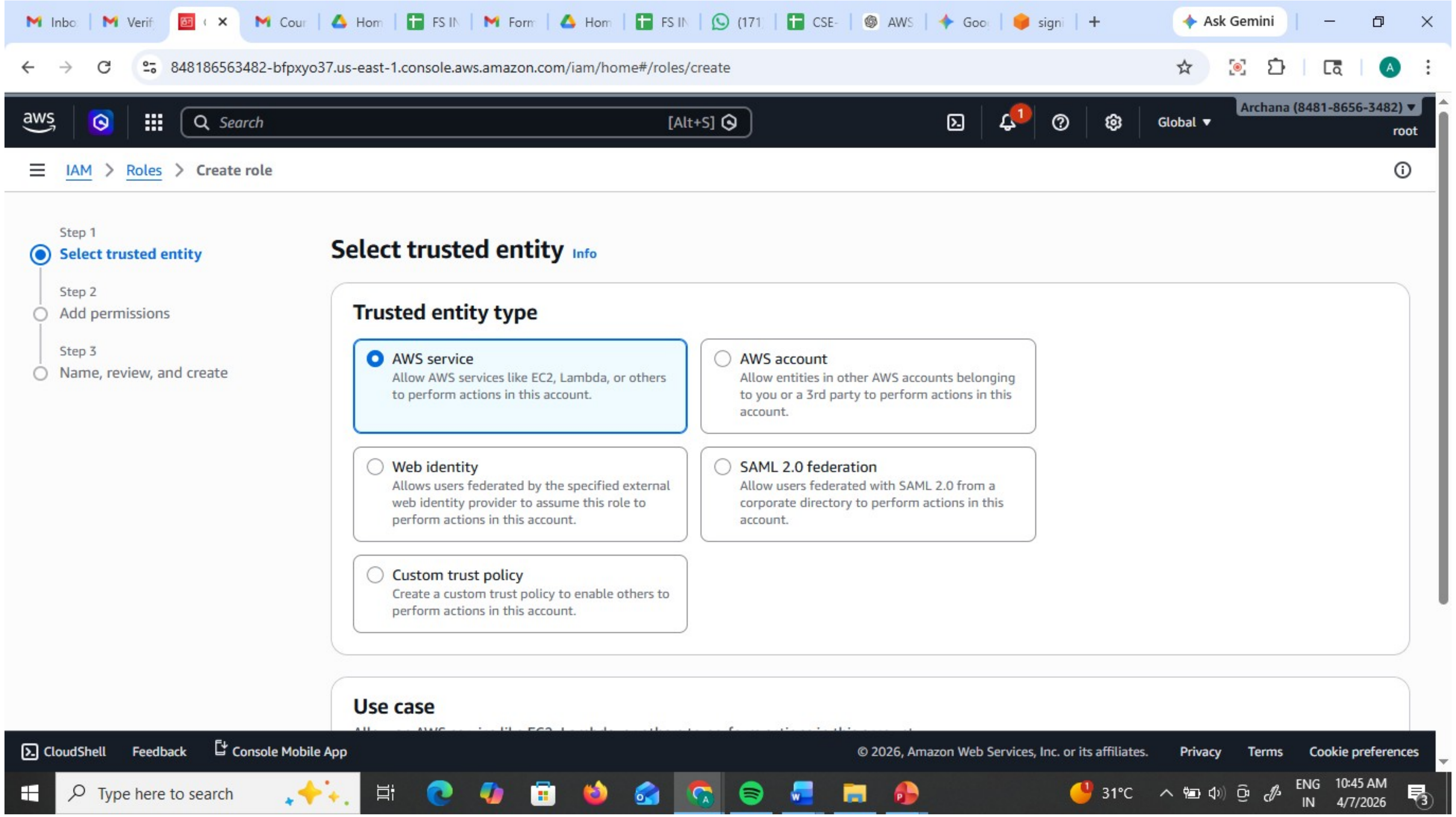
Procedure: Attaching a Role to an EC2 Instance



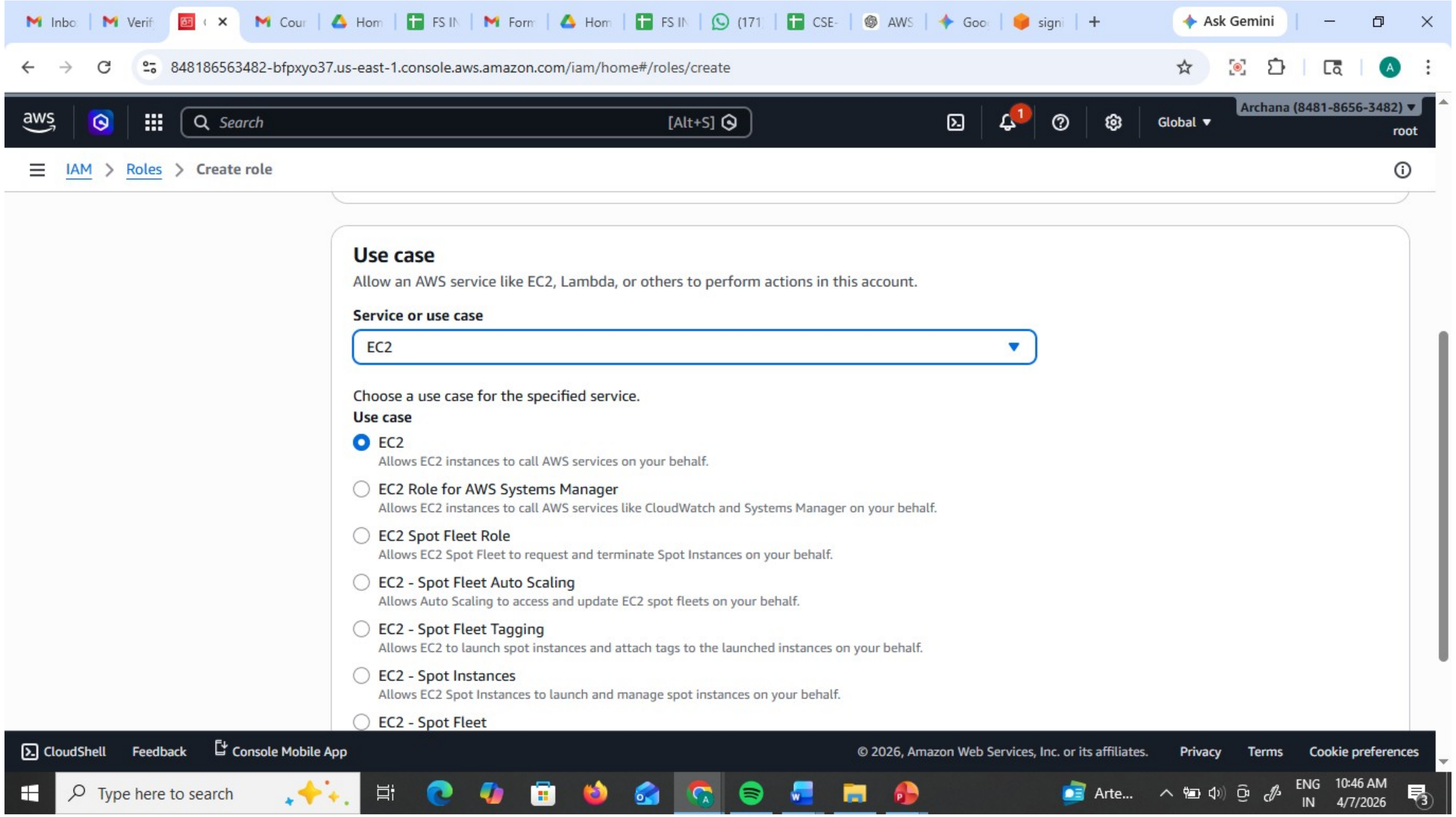
Create the Role: Go to IAM > Roles > Create role



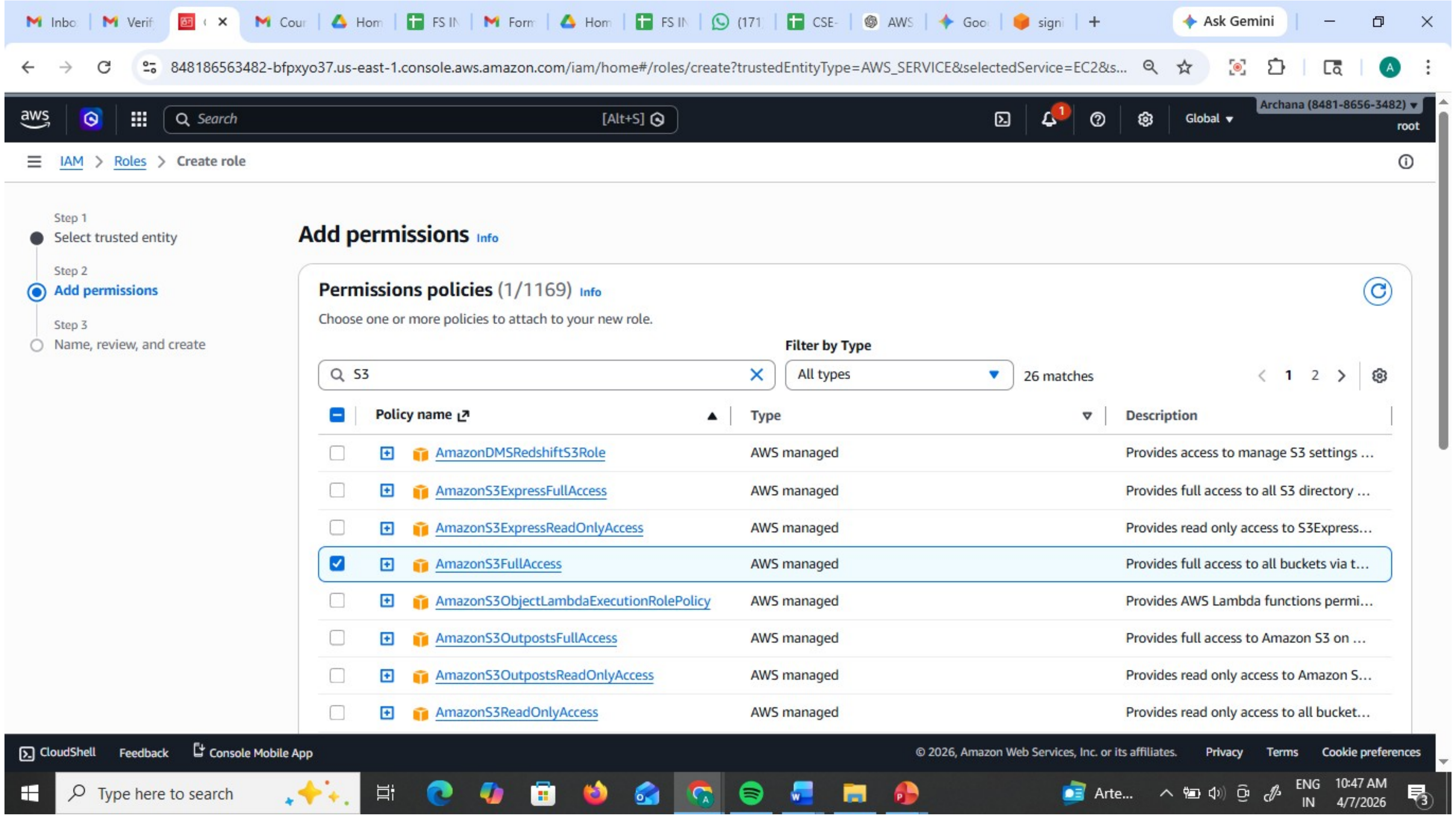
Select Trusted Entity: Choose AWS Service.



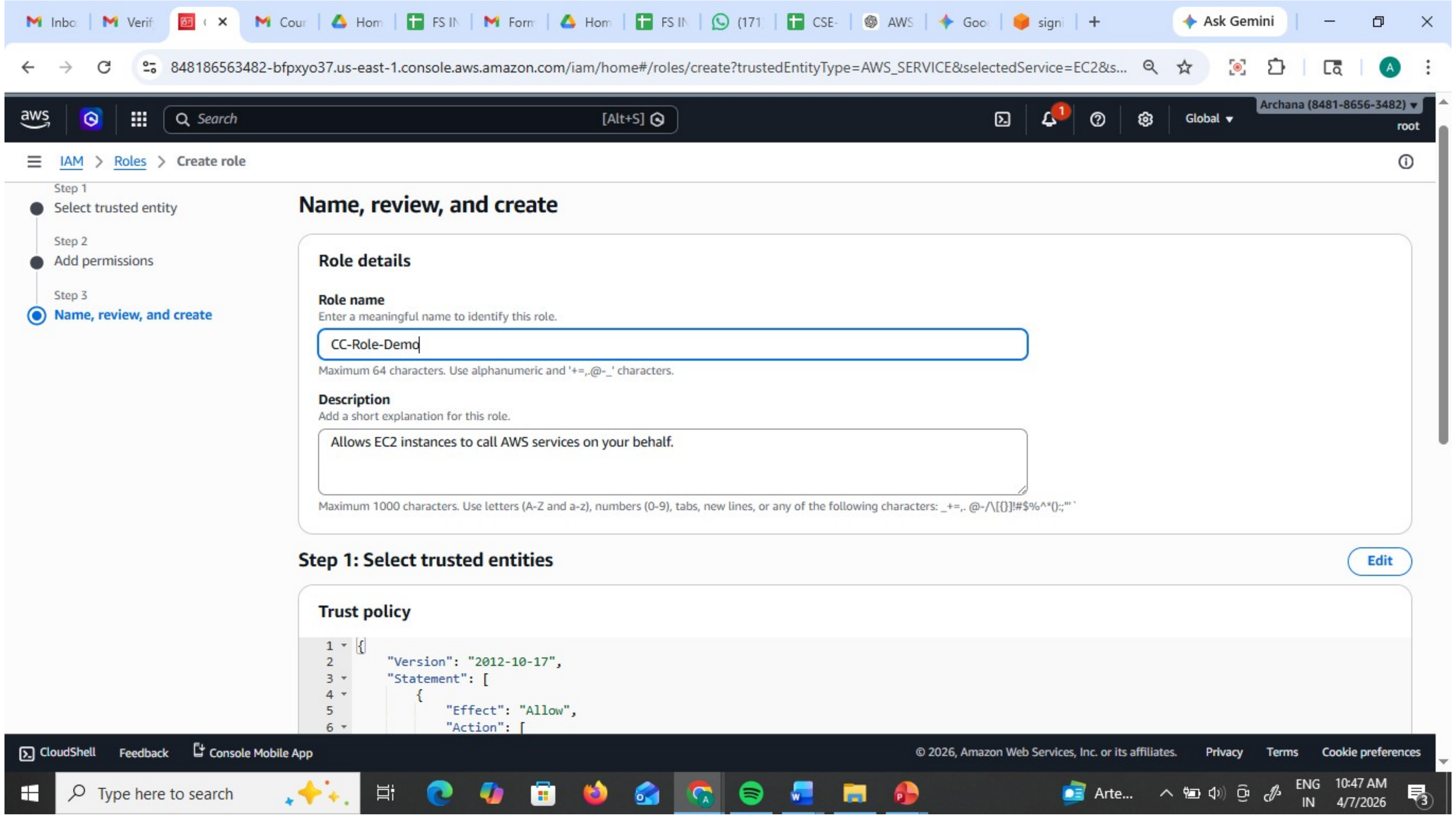
Under the "Service or use case" dropdown, select EC2. (This tells AWS: "I trust the EC2 service to use this role").



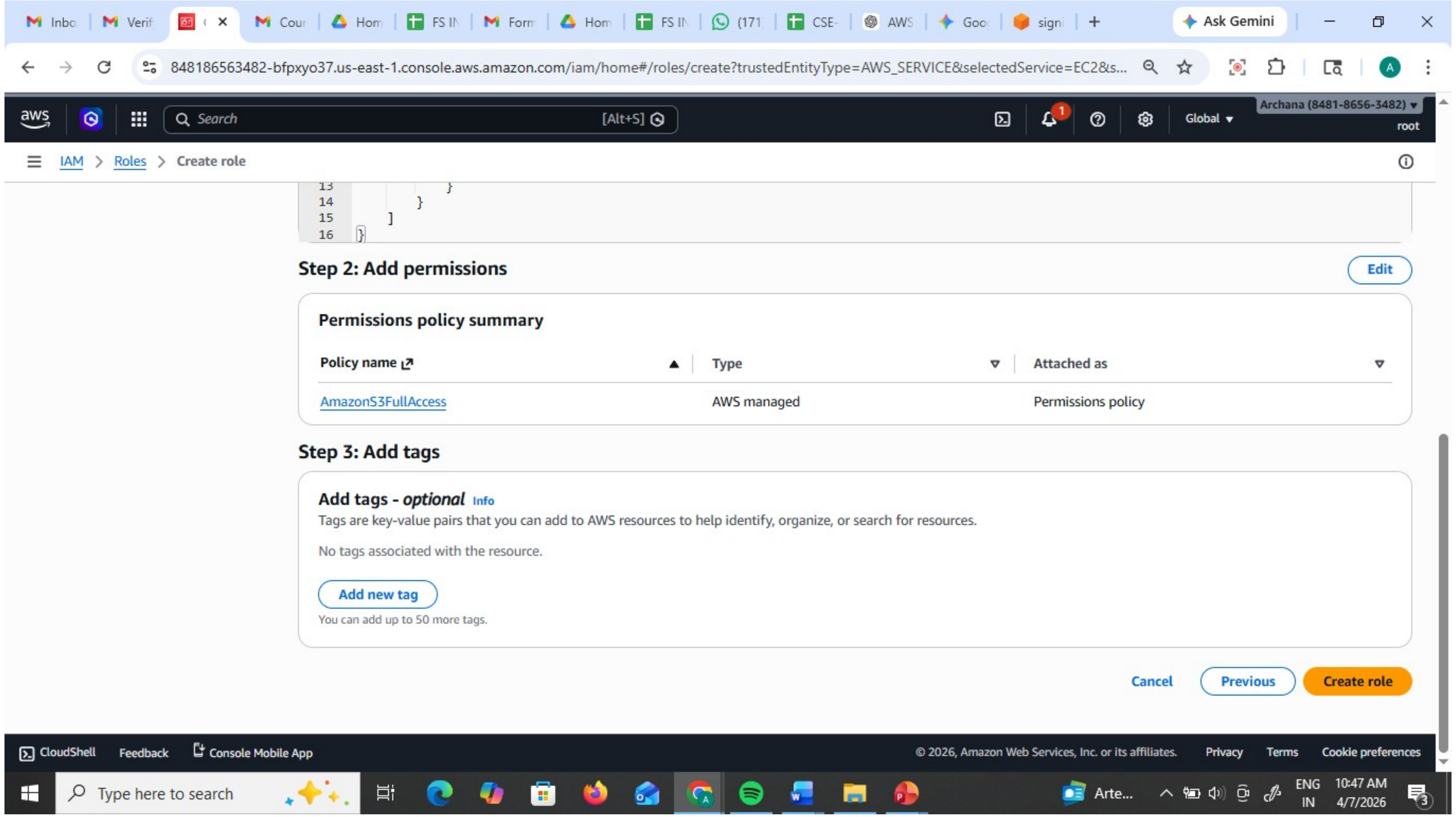
Add Permissions: Search for and check AmazonS3FullAccess.



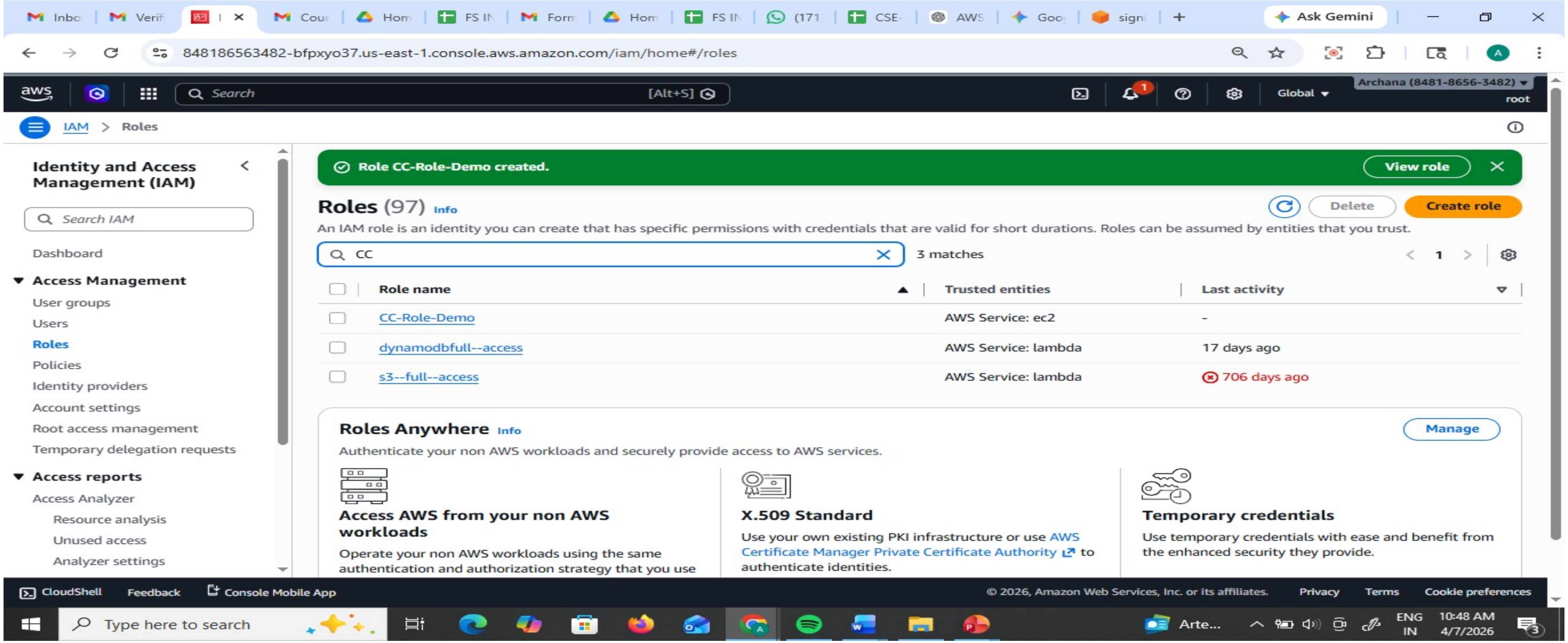
Name & Review: Name it (e.g., EC2-S3-ReadOnly-Role) and click Create role.



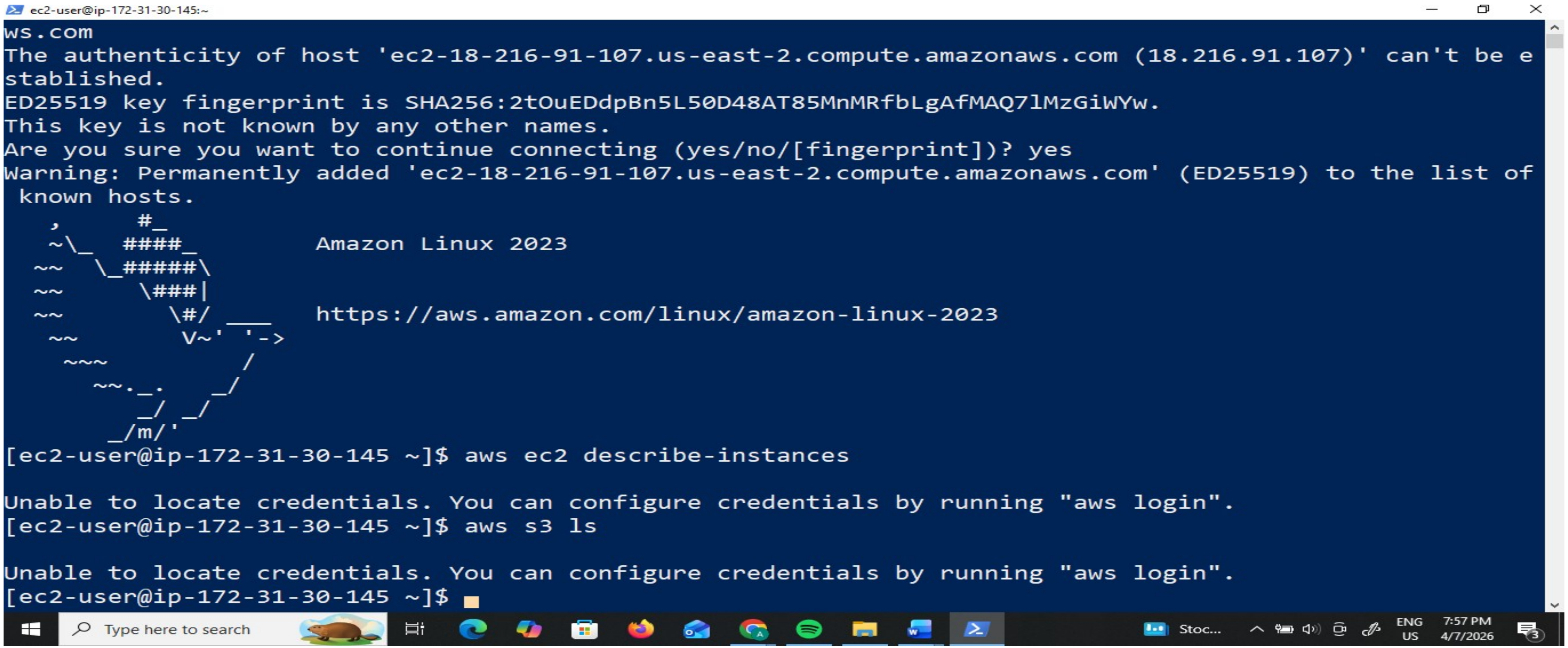
Name & Review: Name it (e.g., EC2-S3-ReadOnly-Role) and click Create role.



Role has been created



Create the ec2 instance and and connect to it and check for aws ec2 describe-instances



Attach to Instance:
Go to the EC2 Dashboard.
Select your running instance.
Click Actions > Security > Modify IAM role.

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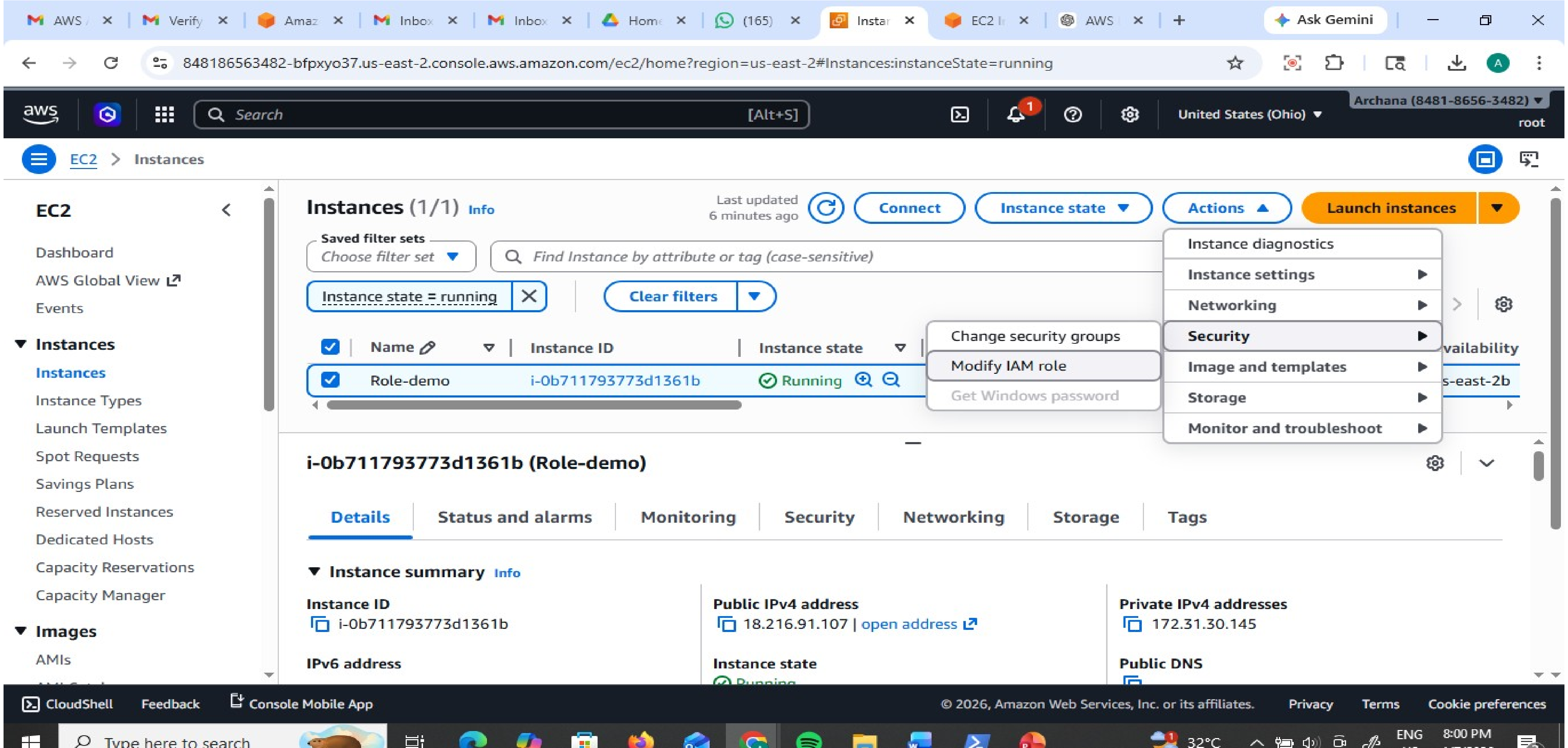
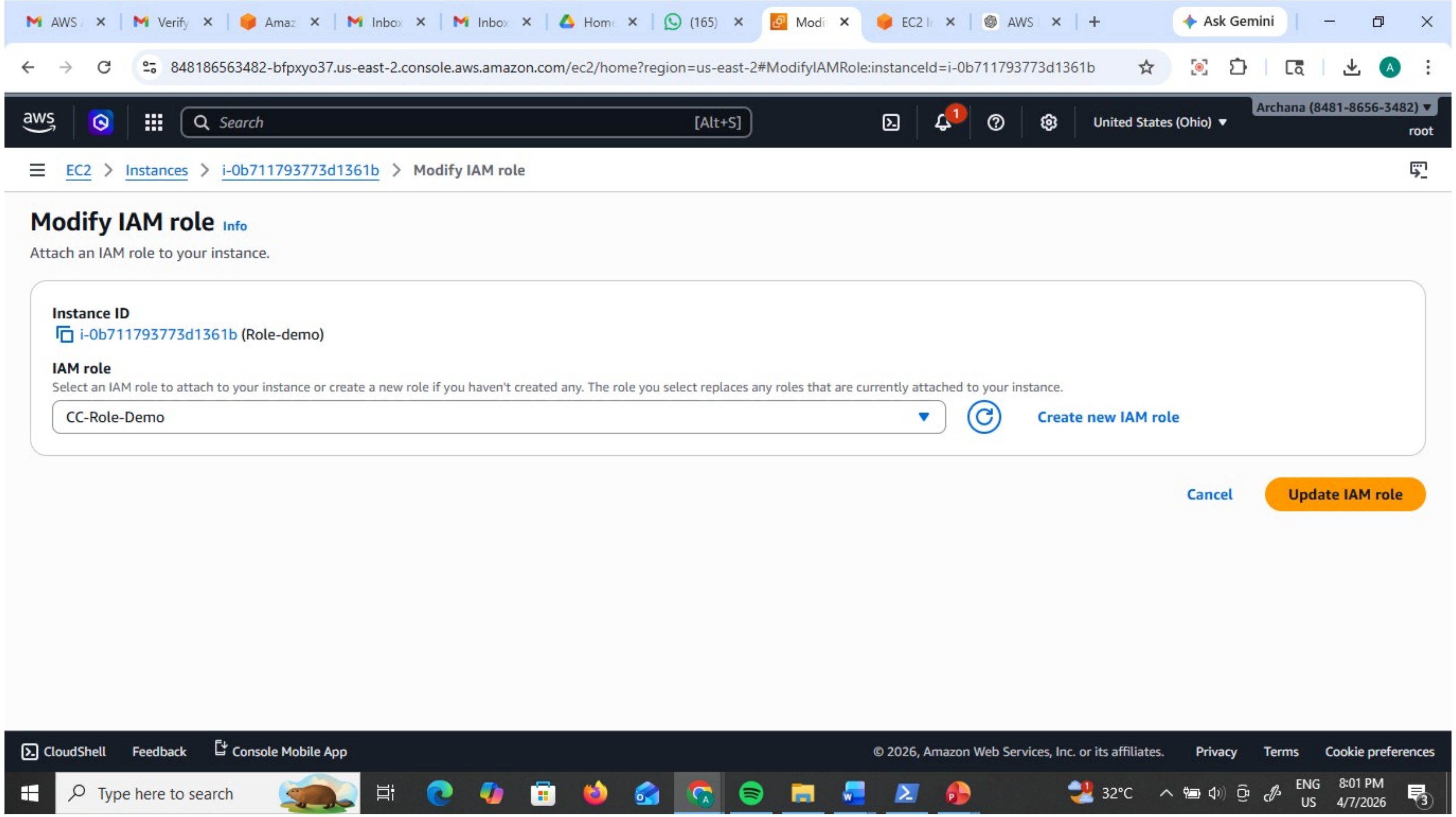


Select the role you just created and click Update IAM role



Verify via CLI:

SSH/Connect into your EC2 instance.

Type aws s3 ls.

The Result: The command works immediately.

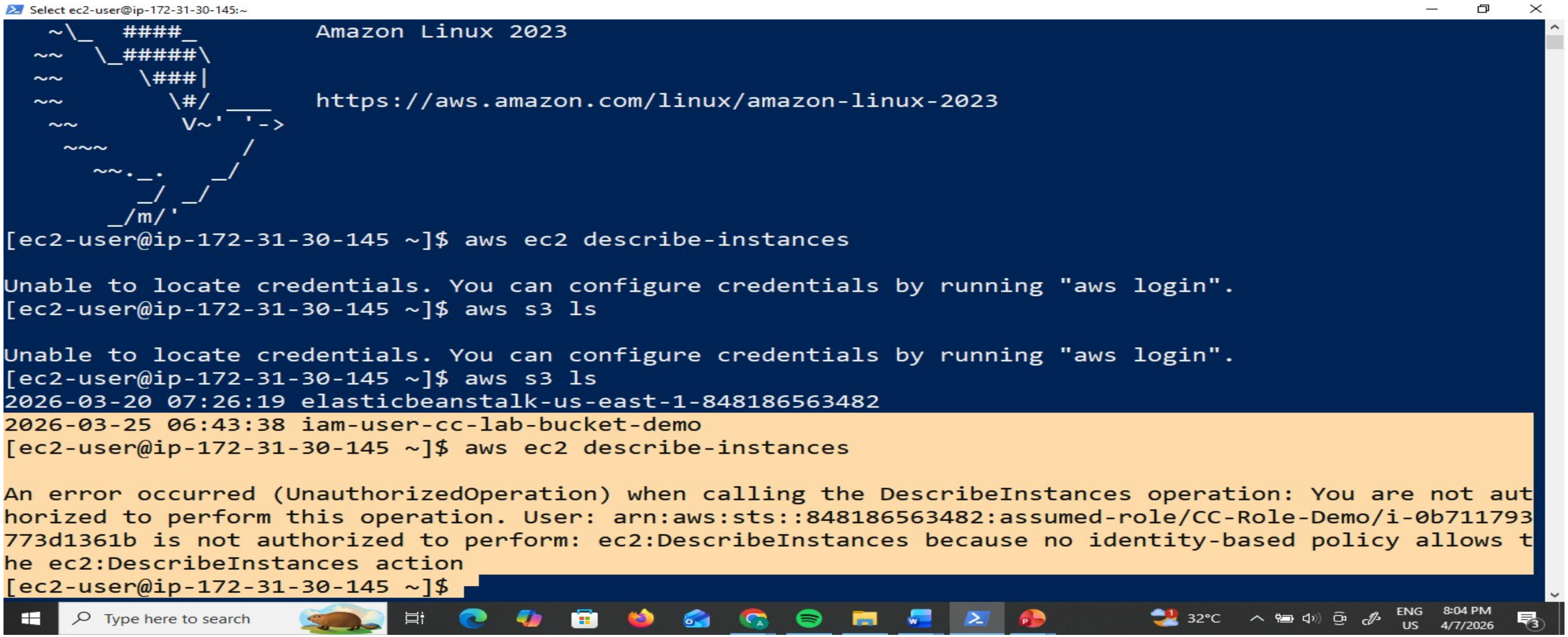
Select ec2-user@ip-172-31-30-145:~

```
ED25519 key fingerprint is SHA256:2tOuEDdpBn5L50D48AT85MnMRfbLgAfMAQ7lMzGiWYw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-18-216-91-107.us-east-2.compute.amazonaws.com' (ED25519) to the list of
 known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-30-145 ~]$ aws ec2 describe-instances

Unable to locate credentials. You can configure credentials by running "aws login".
[ec2-user@ip-172-31-30-145 ~]$ aws s3 ls

Unable to locate credentials. You can configure credentials by running "aws login".
[ec2-user@ip-172-31-30-145 ~]$ aws s3 ls
2026-03-20 07:26:19 elasticbeanstalk-us-east-1-848186563482
2026-03-25 06:43:38 iam-user-cc-lab-bucket-demo
[ec2-user@ip-172-31-30-145 ~]$
```

Type here to search 32°C ENG US 8:03 PM 4/7/2026

We can here role has only s3 full access only here-- aws ec2 describe-instances –unauthorized

You did not have to provide an Access Key or Secret Key. AWS automatically provided Temporary Security Credentials to the instance.



Scenario based questions:

1. Scenario: New Developer Access

You hired a new developer who needs access to EC2 and S3.

Question: How will you provide access?

2. Scenario: Temporary Access Needed

A user needs access only for 2 days. How to do this?

3. Scenario: Access Key Leak

A developer accidentally exposed access keys on GitHub.How to resolve this?

4. Scenario: CLI Access Required

User needs programmatic access. How to do this?

5. Scenario: 50 Employees Same Permissions

50 employees need same S3 access. How to do this?

6. Scenario: Change Permissions for Multiple Users

You need to revoke access from 20 users. How to do this?

7. Scenario: Different Departments

HR, Dev, Admin need different access. How to do this?

8. Scenario: EC2 Access to S3

An EC2 instance needs access to S3 without credentials.How to do this?

9. Scenario: Cross-Account Access

Account A wants to access resources in Account B. How to do this?

10. Scenario: Lambda Access to DynamoDB

Lambda needs to read/write DynamoDB. How to give this?

11. Scenario: Temporary Elevated Access

Admin access needed temporarily. How to do this?

12. Scenario: Restrict S3 Access to Specific Bucket

User should only access one bucket. How to do this?

13. Scenario: Deny Delete Actions

Users should not delete resources. Why this happens?

14. Scenario: Time-Based Access

User should access only during office hours. How to do this?

15. Scenario: IP-Based Access

Allow access only from office network. How to do?

16. Scenario: Secure Best Practice

How do you manage permissions efficiently?

17. Scenario: Root User Usage

When should root user be used?

18. Scenario: Multiple Services Access

User needs EC2 + S3 + DynamoDB. How to do this?

19. Scenario: Audit Permissions

How to track user activity?

20. Scenario: Over-Permission Issue

User has too much permission. How to solve this?